

Project Report

Summary

This project examines operational flight delay patterns across airlines, weekdays, and airports to identify structural inefficiencies within airline networks.

The original dataset consisted of over **10,50,000 flight-level records across 22 operational variables**. To ensure analytical feasibility and maintain dashboard responsiveness within Google Sheets, a statistically randomized subset of **5,000 flights** was generated using Google Colaboratory. All 22 columns were retained to preserve structural richness.

The analysis reveals that flight delays are not randomly distributed. Instead, they are concentrated among specific airlines, peak weekdays, and hub airports. These findings indicate systemic operational bottlenecks rather than isolated scheduling anomalies.

The resulting dashboards provide a structured decision-support framework to monitor performance and guide targeted operational improvements.

Introduction

The aviation sector operates within tightly interconnected scheduling systems. Even minor delays can cascade through networks, resulting in increased operational cost, passenger dissatisfaction, and inefficient resource allocation.

The objective of this analysis is to identify structural delay patterns across airlines, weekdays, and airports, and determine where operational adjustments can reduce systemic inefficiencies.

Dataset Overview

Source and Structure

Source: <https://www.kaggle.com/datasets/usdot/flight-delays>

Type: Raw operational flight records

Dimensions: 10,50,000+ rows and 22 columns

Sampling Methodology

Due to the computational limitations associated with large-scale pivot operations in Google Sheets, random sampling was conducted using Google Colaboratory.

Sampling process:

- 5,000 records selected through unbiased random sampling
- All 22 columns retained
- No filtering based on airline, airport, or delay value

Data Preparation and Cleaning

Raw Data Preparation

Sheet: Raw_Data (Duplicate Removed)

- Imported the randomly sampled dataset (5,000 records) generated using Google Colab.
- Identified and removed duplicate rows to ensure each flight record was unique.
- This sheet became the base reference for all subsequent cleaning steps.

Tail Number Validation

Sheet: Cleaned_Tail_Number (39 Rows Removed)

- Examined the Tail Number column for missing or invalid aircraft identifiers.
- Removed records where tail numbers were blank or incorrectly formatted.
- Ensured all retained records had valid aircraft identifiers suitable for aircraft-level or fleet-level analysis.

Departure Time Cleaning

Sheet: Cleaned_Departure_Time (130 Rows Removed)

- Inspected the Departure Time field for missing, zero, or invalid entries.
- Removed records with unusable time values.
- Standardized the departure time format to enable consistent time-based delay analysis.

This ensured reliability in calculating departure-based KPIs and temporal patterns.

Taxi-Out Time Verification

Sheet: Cleaned_Taxi_Out (2 Rows Removed)

- Reviewed Taxi-Out Time for unrealistic or extreme values.
- Removed records with operationally implausible taxi durations.

This step ensured that ground movement time metrics were reliable for operational benchmarking.

Elapsed Time Validation

Sheet: Cleaned_Elapsed_Time (8 Rows Removed)

- Checked the Elapsed Time column for missing or inconsistent flight duration values.
- Removed records with invalid duration data.

This ensured flight time comparisons and delay benchmarking were based on accurate operational durations.

Final Dataset Filtering

Sheet: Final_Cleaned_Data (Cancelled & Diverted Removed)

Flights marked as:

- Cancelled
- Diverted

were removed from the final dataset.

These records do not represent completed operational cycles and would distort delay-based KPIs if included.

KPI Framework

Airline-level KPIs

- **Total Airlines:** Indicates dataset coverage and competitive benchmarking scope.
- **Total Flights:** Represents valid completed flights used for analysis.
- **Highest Airline Share:** Identifies the airline with the largest traffic share, useful for contextualizing delay contribution.
- **Average Departure Delay per Airline:** Measures scheduling and ground-operation efficiency.
- **Average Arrival Delay per Airline:** Reflects end-to-end operational reliability.
- **Top 5 Airline Performance:** Highlights comparative punctuality leaders and laggards.

Temporal KPIs

- **Average Departure Delay per Day**
- **Average Arrival Delay per Day**

Airport-level KPIs

- **Average Arrival Delay per Airport**
- **Average Departure Delay per Airport**

Analytical Findings

Airline Performance

- Delay levels vary meaningfully across airlines, indicating uneven operational efficiency.
- The airline with the highest traffic share does not necessarily exhibit the highest delay, suggesting performance differences are not purely volume driven.
- A small number of airlines contribute disproportionately to overall delay averages.

Insight:

Operational inefficiencies are carrier-specific and concentrated rather than industry-wide.

Day-of-Week Patterns

- Monday records the highest average delays for both departures and arrivals.
- Midweek performance appears comparatively stable.
- Weekend delays are moderate but consistent.

Insight:

Delay patterns are influenced by structured demand cycles, particularly at the start of the week.

Airport-Level Performance

- Certain airports show sharp spikes in average delay, acting as bottleneck nodes.
- Delay concentration is uneven across the airport network.
- Arrival and departure delays show similar patterns at high-delay airports.

Insight:

Operational congestion at specific airports contributes significantly to network-wide delay propagation.

Dashboard Architecture

1. Executive KPI Layer

- Total Airlines
- Total Flights
- Highest Market Share
- Airline Filter

Purpose:

Provides high-level context and enables interactive segmentation.

2. Comparative Performance Layer

- Average Delay per Airline (Arrival & Departure)
- Top 5 Airline Comparison
- Average Delay per Day

Purpose:

Supports benchmarking across carriers and temporal patterns.

Design:

Bar charts for clarity, consistent scales, and clean comparative layout.

3. Airport Network Layer

- Average Delay per Airport (Arrival & Departure)

Purpose:

Identifies structural bottlenecks and delay concentration nodes.

Design:

Wide airport coverage enabling spike detection and performance comparison.

Strategic Recommendations

1. Carrier-Specific Performance Optimization

Airlines with elevated delay averages should conduct targeted operational reviews focusing on turnaround time efficiency and crew allocation.

2. Peak Weekday Capacity Planning

Given elevated Monday delays, schedule smoothing and buffer allocation should be implemented during high-demand days.

3. Hub Airport Congestion Mitigation

Airports demonstrating persistent delay spikes require ground resource reassessment and slot management refinement.

4. Delay Escalation Monitoring

Introduce threshold-based monitoring systems to flag flights exceeding early delay thresholds to prevent cascading operational disruptions.

5. Data-Driven Continuous Monitoring

Institutionalize dashboard tracking as an operational performance tool rather than a one-time analysis exercise.

Conclusion

This analysis confirms that flight delays are structurally concentrated across specific airlines, weekdays, and airports.

Operational inefficiencies are measurable, clustered, and influenced by traffic patterns and network concentration nodes.

The structured KPI framework and integrated dashboard provide a scalable decision-support system capable of driving continuous operational improvement and performance benchmarking.

The project demonstrates how structured data preparation, systematic KPI design, and layered dashboard architecture can convert raw operational records into actionable aviation performance intelligence.

Contribution Matrix

Role	Primary Responsibility	Role/Owners Name
Project Lead	Overall coordination, milestone tracking, mentor communication, final integration of all components	Akhil Mishra(2401010051)
Data Lead	Sampling in Google Colab, raw data handling, structured cleaning workflow, validation of final dataset	Ankit Kumar Pandey
Analysis Lead	KPI framework design, pivot table logic, statistical interpretation, trend identification	Aryan
Dashboard Lead	Dashboard layout design, visualization logic, filter implementation, usability refinement	Prachee Dhar
PPT & Quality Lead	Presentation design, narrative structuring,	Abhishek Sharma

	formatting consistency, final review for coherence	
Strategy Lead	Insight translation into actionable recommendations, business implications, conclusion framing	Krishna Dubale
Operations Lead	Data integrity oversight, documentation of cleaning logs, version history monitoring, ensuring alignment between analysis and final outputs	Arya Pandey